G	$Z(G)$
abelian group	G
$S_3 \cong D_3$	Z_1
D_4	$\mathbb{Z}_2$
$SO(2k+1, \mathbb{R})$	$\mathbb{Z}_1$
$SO(2k, \mathbb{R})$	$\mathbb{Z}_2$
$U(N)$	$U(1)$
$SU(N)$	$\mathbb{Z}_N$
$Sp(2N, \mathbb{R})$	$\mathbb{Z}_2$

Table 1: A few examples of groups and their centres. The other exotic example are $\mathbb{G}_2$, $\mathbb{F}_4$, $\mathbb{E}_6$, $\mathbb{E}_7$, $\mathbb{E}_8$, which respectively have $\mathbb{Z}_1$, $\mathbb{Z}_1$, $\mathbb{Z}_3$, $\mathbb{Z}_2$, $\mathbb{Z}_1$.

and its subgroup

$$H = \left\{e, a^2, a^4, \ldots, a^{2n-2}\right\},$$

which are both abelian, and consequently normal. We can construct the left cosets

$$eH (= a^2 H = a^4 H = \ldots) = H = \left\{e, a^2, \ldots, a^{2n-2}\right\}$$

and

$$aH (= a^3 H = a^5 H = \ldots) = \left\{a, a^3, \ldots, a^{2n-1}\right\},$$

while the quotient space

$$\mathbb{Z}_{2n}/H = \{eH, aH\}$$

is in fact a group (isomorphic to $\mathbb{Z}_2$), thankfully.[3]

Another interesting example is $G = SU(2)$ and $H = \{\mathbb{I}_2, -\mathbb{I}_2\}$ (where $H \subset G$, clearly). We note that $\forall A \in G$

$$A \cdot (\pm I_2) = \pm A = \pm \mathbb{I}_2 \cdot A,$$

and thus $\forall A \in G$, and $\forall h \in H$

$$A \cdot h \cdot A^{-1} = h \in H.$$

Since H is normal, $SU(2)/H$ is a group, and given that $H \cong \mathbb{Z}_2$

$$SU(2)/\mathbb{Z}_2 \cong SO(3)$$

which is a realisation of the facts we introduced about $SU(2)$ and $SO(3)$. Another interesting thing worth to mention: H contains all elements of G that commute with all $g \in G$. A(n abelian) subgroup with this property is called the *centre* of the group.

7 Lesson 7

7.1 Kernel and image

Going back to the definition of group homomorphisms (functions between groups that preserve the group product) we can defined its kernel and its image.

[3] It is important to note that a^2 can be different from e, but $a^2 H = eH$.

Definition 7.1 (Image)**.** The kernel $\ker\mu$ of an homomorphism $\mu : G_1 \to G_2$ is defined as

$$\ker\mu = \{g_1 \in G_1 : \mu(g_1) = e_2 \in G_2\}$$

where e_2 is the unit element of G_2.

Definition 7.2 (Kernel)**.** The image $\mathrm{Im}\,\mu$ of an homomorphism $\mu : G_1 \to G_2$ is defined as

$$\mathrm{Im}\,\mu = \{g_2 \in G_2 : \exists g_1 \in G_2 : \mu(g_1) = g_2\}.$$

The kernel is a subgroup of G_1, and the image is a subgroup of G_2: it can be easily shown that the group product is closed with respect to these sets.

Theorem 7.3
$G_1/\ker\mu$ is isomorphic to $\mathrm{Im}\,\mu$.

Proof. Let us consider

$$\begin{aligned} i :& G_1/\ker\mu \to \mathrm{Im}\,\mu \\ & g_1K \mapsto i(g_1K) = \mu(g_1), \end{aligned}$$

which is potentially ambiguous: for all $k \in \ker\mu$ $g_1kK = g_1K$.[1]

First, we must show that this definition is well-posed. If we have $g_1K = h_1K$, it necessarily means that

$$h_1 = g_1k_1$$

for some $k_1 \in \ker\mu$, but then

$$\mu(h_1) = \mu(g_1k_1) = \mu(g_1)\mu(k_1) = \mu(g_1)$$

because $\mu(k_1) = e_2$ by definition. Whichever representative of the coset we decide to use, we always get the same result.

We can check explicitly that i is both injective and surjective: i is invertible, and a group homomorphism (because it is defined through the homomorphism μ), therefore it is an isomorphism. □

For instance, we can consider $SU(2)/\mathbb{Z}_2$, where $\mathbb{Z}_2$ is the centre of $SU(2)$.[2] We can actually use this result to show that $SU(2)/\mathbb{Z}_2 \cong SO(3)$.

We define the group homomorphism

$$\mu : SU(2) \to SO(3)$$

such that

$$(\mu(A))_{ab} = \frac{1}{2}\,\mathrm{Tr}\left(\sigma^a A \sigma^b A^{-1}\right)$$

where σ^i are the three Pauli matrices. To prove that this is in fact a group homomorphism we need to check that the group product is preserved

If you have some free time check it

[1] The natural order in which we write the elements of these sets can be different, but in essence they are the same.

[2] We already stated that the centre of $SU(N)$ is always $\mathbb{Z}_N$. We can think of $SU(N)$ as the group of unitary $N \times N$ matrices with $\det = 1$: the centre consist of the group made by the roots of unity (multiplied by $\mathbb{I}$), which is isomorphic to $\mathbb{Z}_N$.

$$(\mu(A)\cdot\mu(B))_{cd} = \sum_{a=1}^{3} (\mu(A))_{ca}(\mu(B))_{ad} \stackrel{?}{=} (\mu(A\cdot B))_{cd}.$$

It is also possible to prove that μ is surjective, but it is not injective because $\mu(A) = \mu(-A)$.

The kernel of μ is $\{\mathbb{I}, -\mathbb{I}\} \cong \mathbb{Z}_2$, because

I don't think this implication works.

$$(\mu(\mathbb{I}))_{ab} = \frac{1}{2}\operatorname{Tr}(\sigma^a\sigma^b) = \frac{1}{2}\operatorname{Tr}(\sigma^b\sigma^a) = \frac{1}{4}\operatorname{Tr}(\sigma^a\sigma^b + \sigma^b\sigma^a) = \frac{1}{4}\operatorname{Tr}\{\sigma^a, \sigma^b\} = \frac{1}{4}\operatorname{Tr}(2\mathbb{I}\delta_{ab}) = \delta_{ab}$$

and thus $\mu(\mathbb{I}_{2\times2}) = \mathbb{I}_{3\times3}$.

By the theorem, it is clear that $SU(2)/\mathbb{Z}_2$ must be isomorphic to $SO(3)$. This fact is relatively interesting because $SU(2)$ is not only the kernel of a random isomorphism, but also the centre of the group.

7.2 Representation theory

We studied groups in a general sense, by only considering the their general properties. When we want to have an "example" of a set with these properties, we obtain a *realisation* of the group, and if this set contains square matrices (aka linear transformation in a vector space) we have a *representation*.

Unfortunately, for historical reason when we work with groups like $SU(2)$ we always think of them in terms of matrices, but in principle it is not needed.

Definition 7.4 (Complex vector space)**.** A complex vector space V is a set in which we have an internal operation (called *sum*), with an identity element $\vec{\mathbf{0}}$, with a further operation: multiplication by a scalar (aka a complex number), satisfying

1. $\mu(\vec{\mathbf{v}}_1 + \vec{\mathbf{v}}_2) = \mu\cdot\vec{\mathbf{v}}_1 + \mu\vec{\mathbf{v}}_2$
2. $(\mu_1+\mu_2)\vec{\mathbf{v}} = \mu_1\vec{\mathbf{v}} + \mu_2\vec{\mathbf{v}}$
3. $\mu_1\cdot(\mu_2\vec{\mathbf{v}}) = (\mu_1\cdot\mu_2)\vec{\mathbf{v}}$
4. $1\cdot\vec{\mathbf{v}} = \vec{\mathbf{v}}$ and $0\cdot\vec{\mathbf{v}} = \vec{\mathbf{0}}$.

A linear combination of vectors is simply defined as

$$\sum_{i=1}^{k} \mu_i\cdot\vec{\mathbf{v}}_i$$

and we say that a set of vectors $\{\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \vec{\mathbf{v}}_3, \ldots\}$ is *linearly independent* if

$$\sum_{i=1}^{k} \mu_i\vec{\mathbf{v}}_i = \vec{\mathbf{0}} \iff \mu_i = 0, \forall i.$$

The *dimension* of V is the largest number of independent vectors that can be taken in the same set. And, a *basis* of vectors in V is a set of $n = \dim V$ linearly independent vectors. Given a basis, $\forall v \in V$, we can write

$$\vec{\mathbf{v}} = \sum_{i=1}^{n} v^i\vec{\mathbf{e}}_i$$

where $\{\vec{\mathbf{e}}_i\}$ is the basis, and $\{v^i\}$ the *components* of the vector $\vec{\mathbf{v}}$.

Definition 7.5 (Linear map)**.** Given two vector spaces V_1 and V_2, the function $L : V_1 \to V_2$ is called a linear map when

$$L(\mu_1 \vec{\mathbf{v}}_1 + \mu_2 \vec{\mathbf{v}}_2) = \mu_1 L(\vec{\mathbf{v}}_1) + \mu_2 L(\vec{\mathbf{v}}_2)$$

for all $\mu_1, \mu_2 \in \mathbb{C}$ and $\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2 \in V_1$.

If a linear function is also invertible, then it is said to be a *vector space isomorphism*: every vector space can be seen as a group (with the sum operation), if we take $\mu_1 = \mu_2 = 1$ the linear map is an homomorphism between groups. If two vector space are isomorphic, it follows that $\dim V_1 = \dim V_2$.

Definition 7.6 (Automorphism)**.** An isomorphism between a vector space and itself is called an automorphism. The set of all automorphisms from V to V is denoted by $\mathrm{Aut}(V)$.

Remark 7.7. The set of automorphisms $\mathrm{Aut}(V)$ is actually a group, if we take the group product as function composition.

Given a linear transformation $L : V_1 \to V_2$, its *kernel* and its *image* are defined as

$$\ker L = \left\{ \vec{\mathbf{v}}_1 \in V_1 : L(\vec{\mathbf{v}}_1) = \vec{\mathbf{0}} 2 \in V_2 \right\} \subseteq V_1$$

and

$$\mathrm{Im}\, L = \{ \vec{\mathbf{v}}_2 \in V_2 : \exists \vec{\mathbf{v}}_1 \in V_1 : L(\vec{\mathbf{v}}_1) = \vec{\mathbf{v}}_2 \} \subseteq V_2.$$

In fact, both the kernel and the image are vector spaces (under the same operations of V_1 and V_2). Moreover, it is possible to prove that

$$\dim(\ker L) + \dim(\mathrm{Im}\, L) = \dim V_1,$$

and also $L : V \to V$ is an automorphism if and only if $\left\{ \vec{\mathbf{0}} \right\} = \ker L$.[3]

Proposition 7.8

Every linear map $L : V_1 \to V_2$, where $n_1 = \dim V_1$ and $n_2 = \dim V_2$, can be fully determined by $n_1 \cdot n_2$ complex numbers: the components of the vectors (of V_2) obtained by applying L to the elements of a basis of V_1.

Proof. If we take a basis of V_1 and a basis of V_2, respectively

$$\{ \vec{\mathbf{e}}_1, \dots, \vec{\mathbf{e}}_{n_1} \} \qquad \left\{ \vec{\mathbf{f}}_1, \dots, \vec{\mathbf{f}}_{n_2} \right\},$$

then

$$L(\vec{\mathbf{e}}_i) = \sum_{j=1}^{n_2} L_{ij} \vec{\mathbf{f}}_j.$$

For an arbitrary $\vec{\mathbf{v}} \in V_1$ that can be written as

$$\vec{\mathbf{v}} = \sum_{i=1}^{n_1} v^i \vec{\mathbf{e}}_i$$

[3] If this is the case then $\dim(\mathrm{Im}\, L) = \dim V$, and thus $\mathrm{Im}\, L = V$.

we can always determine

$$L(\vec{\mathbf{v}}) = L\left(\sum_{i=1}^{n_1} v^i \vec{\mathbf{e}}_i\right) = \sum_{i=1}^{n_1} v^i \sum_{j=1}^{n_2} L_{ij}\vec{\mathbf{f}}_j$$

using only the coefficients L_{ij}. □

You can try to fix the indexes of L_{ij}: j higher right and i lower left? Covectors transform with the inverse matrix, idk

Remark 7.9. When $\dim V_1 = \dim V_2$, the L_{ij} form a square matrix, and when $L \in \mathrm{Aut}(V)$ the matrix is also invertible, therefore $\mathrm{Aut}(V) \cong GL(n, \mathbb{C})$.

Definition 7.10 (Linear representation)**.** The linear representation of a group D is defined as the group homomorphism $D : G \to \mathrm{Aut}(V)$. The *dimension* of this representation is defined as the dimension of V.

Definition 7.11 (Faithful representation)**.** A representation D is said to be faithful when it is injective

$$D(g_1) = D(g_2) \implies g_1 = g_2,$$

and a necessary condition for this is that $\ker D = \{e\}$ where $e \in G$.

Indeed, as we anticipated, giving a representation means choosing a way to represent the element of the group as square matrices (aka automorphisms of V).

Definition 7.12 (Intertwining operator)**.** Given a representation $D_1 : G \to \mathrm{Aut}(V_1)$ and another $D_2 : G \to \mathrm{Aut}(V_2)$, we say that a linear map $A : V_1 \to V_2$ is an intertwining operator when it satisfies

$$AD_1(g) = D_2(g)A, \qquad \forall g \in G.$$

Remark 7.13. If $\dim V_1 = \dim V_2$, A can be an isomorphism (since it is both linear and invertible), then we could write

$$D_2(g) = AD_1(g)A^{-1}$$

and in this case we say that D_1 and D_2 are *equivalent* to each other, and that A is a *similarity transformation*.

Definition 7.14 (Scalar product)**.** A scalar product is a "map" $\langle \vec{\mathbf{v}}_1 | \vec{\mathbf{v}}_2 \rangle$ from $V \times V$ to $\mathbb{C}$ satisfying the following properties

1. $\langle \vec{\mathbf{v}}_1 | \mu_1 \vec{\mathbf{w}}_1 + \mu_2 \vec{\mathbf{w}}_2 \rangle = \mu_1 \langle \vec{\mathbf{v}}_1 | \vec{\mathbf{w}}_1 \rangle + \mu_2 \langle \vec{\mathbf{v}}_1 | \vec{\mathbf{w}}_2 \rangle$,
2. $\langle \vec{\mathbf{v}} | \vec{\mathbf{w}} \rangle = (\langle \vec{\mathbf{w}} | \vec{\mathbf{v}} \rangle)^*$,
3. $\langle \vec{\mathbf{v}} | \vec{\mathbf{v}} \rangle = 0$ if and only if $\vec{\mathbf{v}} = \vec{\mathbf{0}}$.

Using a scalar product, it is possible to define the concept of an *orthonormal basis*, a basis such that $\langle \vec{\mathbf{e}}_i | \vec{\mathbf{e}}_j \rangle = \delta_{ij}$.

Definition 7.15 (Adjoint map)**.** Given a linear function $A : V \to V$, we define the adjoint map $A^\dagger : V \to V$, which satisfies the property

$$\left\langle \vec{\mathbf{v}} \middle| A^\dagger \vec{\mathbf{w}} \right\rangle = \langle A\vec{\mathbf{v}} | \vec{\mathbf{w}} \rangle .$$

Definition 7.16 (Unitary map)**.** A linear map $U : V \to V$ is called unitary when

$$\langle U\vec{\mathbf{v}}|U\vec{\mathbf{w}}\rangle = \langle \vec{\mathbf{v}}|\vec{\mathbf{w}}\rangle .$$

Remark 7.17. A map is unitary if and only if its adjoint map is equal to its inverse.

Prove it!

The unitary maps are a subgroup $\mathrm{Unit}(V)$ of $\mathrm{Aut}(V)$: they are linear, and they are invertible (since $U^\dagger U = UU^\dagger = \mathbb{I}$). A *unitary representation* of a group G is a representation $D : G \to \mathrm{Unit}(V)$: they are important in QM, and elementary particles physics.

In QM, states are represented by vectors $|\psi\rangle \in \mathcal{H}$. In the Hilbert space $\mathcal{H}$, there is a scalar product that has a physical interpretation: $\langle\psi|\psi\rangle$ is proportional to the probability density.

Interestingly, we could redefine $|\psi\rangle \to e^{i\alpha}|\psi\rangle$: scalar product would not change, and for this reason they can represent an observable.[4] In general, if a system is invariant under a group G of transformations and U is a unitary representation of G, then

$$\langle U\psi|U\psi\rangle = \langle\psi|\psi\rangle ,$$

and $|U(g)\psi\rangle$ must obey the same Schrödinger equation as $|\psi\rangle$, and thus

$$[U(g), H] = 0,$$

or equivalently $U(g)GU^{-1}(g) = H$.

If we consider the eigenstates of the Hamiltonian

$$H|\phi_n\rangle = E_n|\phi_n\rangle ,$$

this symmetry can help us resolve their (possible) degeneracy. Linear combinations of degenerate eigenstates form subspaces $\mathcal{H}_n \subset \mathcal{H}$, and clearly

$$U(g)|\phi_n\rangle \in \mathcal{H}_n$$

because $U(g)$ commutes with H.

Studying the dimension of the representations of the symmetries, constrains the multiplicities of the degenerate subspaces.

He was not clear.

[4] For this reason, technically, states are represented by *rays* (aka vectors up to a phase) and not vectors.